

A misbehavior detection system to detect novel position falsification attacks in the Internet of Vehicles

Harun Surej Ilango^a, Maode Ma^{b,*}, Rong Su^a

^a School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore

^b College of Engineering, Qatar University, Qatar

ARTICLE INFO

Keywords:

Internet of Vehicles
Position falsification attacks
Machine learning
Novel attack detection
Basic Safety Messages (BSMs)
VeReMi dataset
Vehicle-to-Vehicle (V2V) communication
Anomaly detection

ABSTRACT

In the Internet of Vehicle (IoV) networks, vehicles exchange periodic Basic Safety Messages (BSMs) containing information regarding speed and position. Safety-critical applications like blind-spot warning and lane change warning systems use the BSMs to ensure the safety of road users. To create chaos in the network, an insider attacker may inject false information into the BSM and broadcast it to nearby vehicles. One such attack is the position falsification attack, where the attacker inserts false information regarding the position in the BSMs. The literature has explored the use of Misbehavior Detection Systems (MDSs) to detect such attacks. But the limitations of the existing approaches are that they either perform exceptionally well in specific environmental settings or have compromised detection accuracy favoring a generalized model. Moreover, all the current machine learning-based detection models are signature-based, which requires prior knowledge about the attacks for effective detection. Motivated by the research gap, we propose a Novel Position Falsification Attack Detection System for the Internet of Vehicles (NPFADS for the IoV) to learn and detect novel position falsification attacks emerging in IoV networks. The performance of NPFADS is analyzed using the metrics precision, recall, F1 score, ROC, and PR curves. The Vehicular Reference Misbehavior (VeReMi) dataset is used as the benchmark for the study. The system's performance is compared to existing MDSs in the literature. The analysis shows that our proposed system outperforms the existing supervised learning models even when initialized with zero knowledge about the novel position falsification attacks.

1. Introduction

Vehicles manufactured these days are equipped with onboard computers and sensors that enable driver-assist technologies such as lane following assist, cruise control, and automatic braking system to improve the safety of road users. The paradigm shift from the concept of connected computers to a network of 'things' revolutionized the digital world and created a fresh wave of development (Adat and Gupta, 2018; Kamal et al., 2021). One such revolutionary development is the penetration of the Internet of Things (IoT) into the transportation sector, thereby realizing the concept of the Internet of Vehicles (IoV). IoV improved the safety of road users further by enabling non-line of sight communications and sensing capabilities in vehicles (Sharma and Liu, 2021; Xu et al., 2022). But the open nature of IoT has brought with

it security vulnerabilities (Katiyar et al., 2013). In IoV, vehicles communicate using the Dedicated Short-Range Communications (DSRC) technique. Each vehicle broadcasts its speed, position, and acceleration to nearby vehicles several times a second through Basic Safety Messages (BSMs) using DSRC. Safety-critical applications such as blind-spot collision warning, lane change warning, and intersection movement assist in using the BSMs to realize these applications. Due to the time-sensitive nature of these applications, the information transmitted in the BSMs is typically unencrypted (Sharma and Jaekel, 2022). In IoV, a vehicle joins the network using the network's public key. Before transmitting a BSM, a legitimate sender digitally signs the BSM to ensure that the messages' contents are not altered. However, a legitimate vehicle could launch an insider attack by inserting false information in the digitally signed BSM. The vehicles receiving the BSMs would not find anything

Abbreviations: AE, AutoEncoder; AMF, Access and Mobility Management Function; AUC, Area Under the Curve; BSM, Basic Safety Message; CA, Certificate Authority; CCR, Correct Classification Rate; DSRC, Dedicated Short Range Communications; FN-BSMD, Fog Node - Basic Safety Message Database; IoV, Internet of Vehicles; MCR, Misclassification Rate; MDS, Misbehavior Detection System; MSE, Mean Squared Error; MVD, Misbehaving Vehicles Database; NADM, Novel Attack Detection Module; NASEA, Novel Attack Sample Extraction Ability; NPFADS for the IoV, Novel Position Falsification Attack Detection System for the Internet of Vehicles; OBU, OnBoard Unit; OBU-BSMD, OnBoard Unit - Basic Safety Message Database; RF, Random Forest; ROC, Receiver Operating Characteristic; RSSI, Received Signal Strength Indicator; RSU, Road Side Unit; V2I, Vehicle-to-Infrastructure; V2V, Vehicle-to-Vehicle; VeReMi, Vehicular Reference Misbehavior

* Corresponding author.

E-mail addresses: harunsur001@e.ntu.edu.sg (H.S. Ilango), mamaode@qu.edu.qa (M. Ma), rsu@ntu.edu.sg (R. Su).

<https://doi.org/10.1016/j.engappai.2022.105380>

Received 30 April 2022; Received in revised form 7 August 2022; Accepted 23 August 2022

Available online 13 September 2022

0952-1976/© 2022 Elsevier Ltd. All rights reserved.

suspicious as the messages have come from a legal vehicle. Therefore, the efficacy of such cryptographic techniques dwindles in the presence of an insider attacker in the network. One such insider attack is the position falsification attack, in which the attacker broadcasts incorrect information regarding its position. The vehicles receiving these messages believe that the BSMs are true and adjust their speed and acceleration accordingly, leading to accidents. Hence, it is of cardinal importance to protect the network from these attacks.

The literature has explored the use of Misbehavior Detection Systems (MDSs) to protect the network from position falsification attacks. The two approaches used by a MDS to detect position falsification are (i) entity-centric — identifies misbehaving vehicles in the network and (ii) data-centric — detects false information shared among vehicles. [Sharma and Liu \(2021\)](#) proposed a data-centric machine learning-based MDS to detect position falsification attacks. The performance of the machine learning models — Support Vector Machine, K-Nearest Neighbor, Naïve Bayes, Random Forest, Ensemble Boosting, and Ensemble Voting, were analyzed using the Vehicular Reference Misbehavior (VeReMi) Dataset ([van der Heijden et al., 2018](#)). Additional features of vehicle mobility, viz., location plausibility and movement plausibility, were derived from the original dataset to improve the detection accuracy. The empirical analysis showed that the ensemble methods performed the best overall. However, a deeper analysis on the detection ability of the individual attacks showed that the ensemble methods hardly detect constant offset attacks. Similarly, [Sharma and Jaekel \(2022\)](#) proposed a MDS that achieved better attack detection performance than ([Sharma and Liu, 2021](#)). The proposed MDS was deployed on Road Side Units (RSUs) that detected and flagged misbehaving vehicles. Information regarding the flagged vehicles was broadcasted to the vehicles in the network. The RSUs used a data-centric approach to detect misbehaving vehicles in the network, whereas the OnBoard Units (OBUs) followed an entity-centric approach. The major shortcoming of the proposed system is that it requires a constant connection to the nearby RSU for effective attack detection. The performance of this system in the absence of connection to a local RSU was unexplored. The RSUs were also required to maintain a shared database of BSMs received from all the vehicles in the network. However, in a large network like IoV, where millions of vehicles communicate at a given time, the size of the shared database would quickly become quite large. This would lead to an increased data retrieval time which inadvertently leads to an increased attack detection time thereby flawing the authors' assumption of negligible communication and database retrieval delays.

[Ercan et al. \(2022\)](#) proposed a data-centric MDS that relied on three additional features of vehicle mobility for effective attack detection: (i) Angle of Arrival (AoA), (ii) the estimated distance between sender and receiver, and (iii) the difference between the declared distance and the estimated distance. The authors claimed that their proposed additional features improve the detection accuracy of position falsification attacks. The feature, estimated distance between sender and receiver, was computed using the measured Received Signal Strength Indicator (RSSI) value at the receiver. Similarly, [So et al. \(2019\)](#) proposed a MDS that used RSSI values at the receiver for attack detection. The MDS detected position falsification attacks by constructing a distance versus RSSI distribution by observing the behavior of benign vehicles. Confidence intervals for RSSI values at a given distance were computed. The sender was classified as malicious if the RSSI value for the received BSM does not fall within the confidence interval for a given distance. The major shortcoming of the MDSs proposed in [Ercan et al. \(2022\)](#), [So et al. \(2019\)](#) is that they relied on the RSSI values for attack detection. The characteristics of a propagation channel change with respect to the environment through which the signal propagates. Therefore, in [Ercan et al. \(2022\)](#), the estimation of the sender's distance by the receiver might be incorrect, leading to a reduction in the detection performance. Similarly, in [So et al. \(2019\)](#), the model required access to the local RSSI distribution for effective attack detection. Developing such RSSI distribution data for all the

different localities where the MDS is deployed is a challenging task. The major shortcoming of all the above-discussed machine learning-based MDSs are that they are signature-based, which requires prior knowledge about the attacks for effective detection. In conclusion, all the above existing solutions in literature have their shortcomings. They either perform exceptionally well in specific environmental settings, i.e., the detection models are tailored for a specific environment, or have compromised detection accuracy favoring a generalized model. Moreover, all the current machine learning based detection models are signature-based, which requires prior knowledge about the attacks for effective detection. Motivated by the research gaps, we propose an AI-based misbehavior detection system to detect position falsification attacks in an IoV environment.

The significant contribution of this paper is our proposed Novel Position Falsification Attack Detection System for the Internet of Vehicles (NPFADS for the IoV) which holds the following outstanding features: (i) The system can detect insider attacks in an IoV network. (ii) The system does not require prior knowledge about the attacks for effective detection. (iii) The system can automatically learn and detect emerging new position falsification attacks in the network. (iv) The system can theoretically learn and detect an indefinite number of position falsification attacks. (v) The system's performance in detecting position falsification attacks is on par with the existing solutions in the literature, even when the system initially has zero knowledge about the novel attacks.

The rest of the paper is organized as follows. Section 2 describes the IoV model and the threat model. The dataset and its associated pre-processing methodology are discussed in Section 3. Section 4 discusses the proposed NPFADS, and Section 5 analyzes the performance of NPFADS. The paper is then concluded in Section 6.

2. The IoV model and the threat model

2.1. The IoV model

[Fig. 1](#) shows the IoV model under study. The vehicles exchange information through Vehicle-to-Vehicle (V2V) communication using the DSRC technique, which has a data rate of 6 Mbps with a communication range of up to 1000 m. Vehicle-to-Infrastructure (V2I) communication uses the 5th Generation (5G) cellular networks. More precisely, FR1 bands (410 MHz–7125 MHz) specified under 5G release 16 by 3GPP are used as they provide a wide area coverage (2 km–30 km) while promising data rates up to 1400 Mbps.

To facilitate communications, an OBU is installed in each vehicle which has two communication interfaces — 5G wireless and DSRC. RSUs are placed alongside the roads as the communication infrastructure is supported by 5G wireless communication. The RSUs are implemented by gNodeBs, and the Access and Mobility Management Function (AMF) in the 5G core network is equipped with a dedicated fog server acting as a fog node. The 5G core interconnects the fog nodes.

2.2. The threat model

A vehicle is registered with the Certificate Authority (CA) to obtain the credentials required to join the IoV network. The authorized vehicles join the network using the credentials and begin communicating with the RSUs and other vehicles. The vehicles use digitally signed BSMs to communicate with each other in the network. To carry out a position falsification attack, an insider attacker holding valid credentials will insert incorrect position information in the BSMs and transmit it to other vehicles. [Fig. 2](#) demonstrates such a scenario where an insider attacker causes an accident by transmitting incorrect position information. The vehicle colored in orange is the insider attacker who broadcasts incorrect position information. The insider attacker is actually at P1 (position 1) but broadcasts its position as P2 (position 2). The safety-critical applications installed in the benign vehicle (colored in green) falsely believe that the road is clear, leading to an accident.

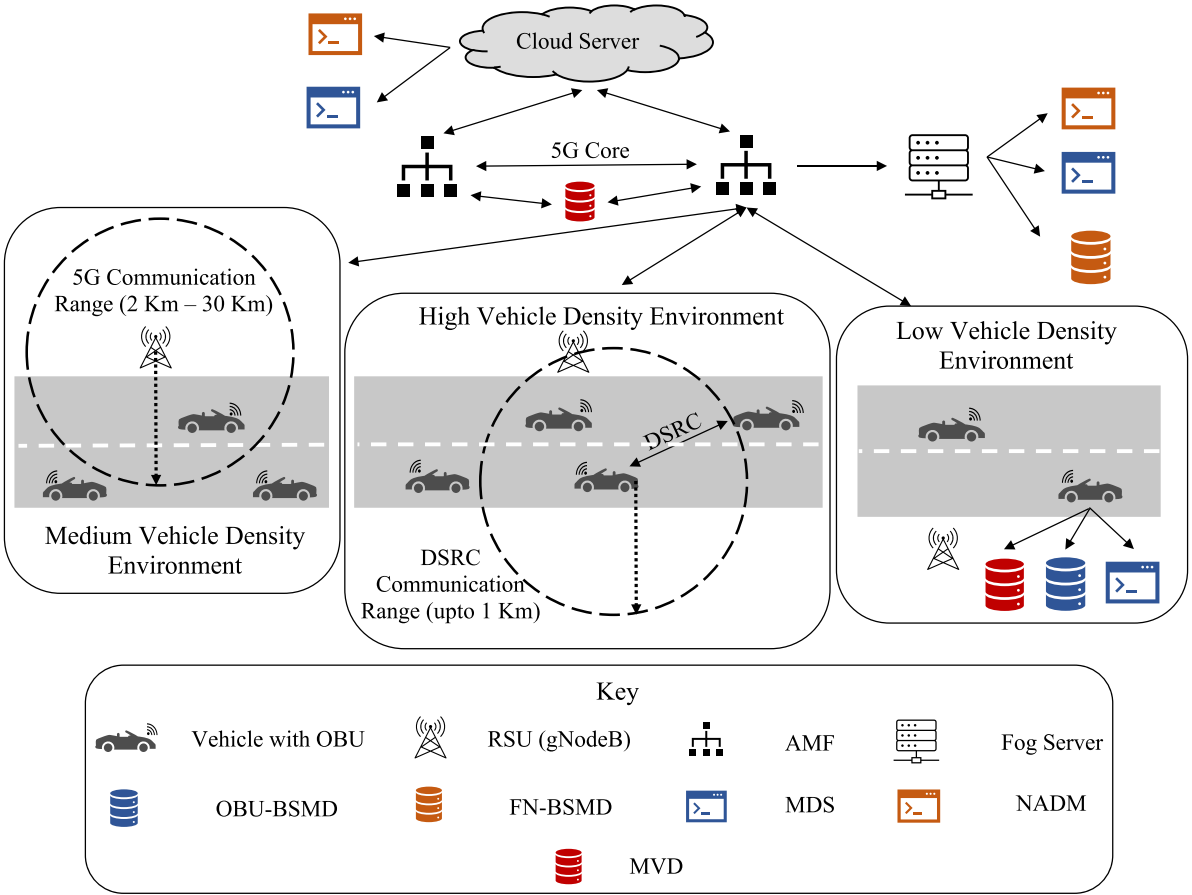


Fig. 1. The IoV model under study.

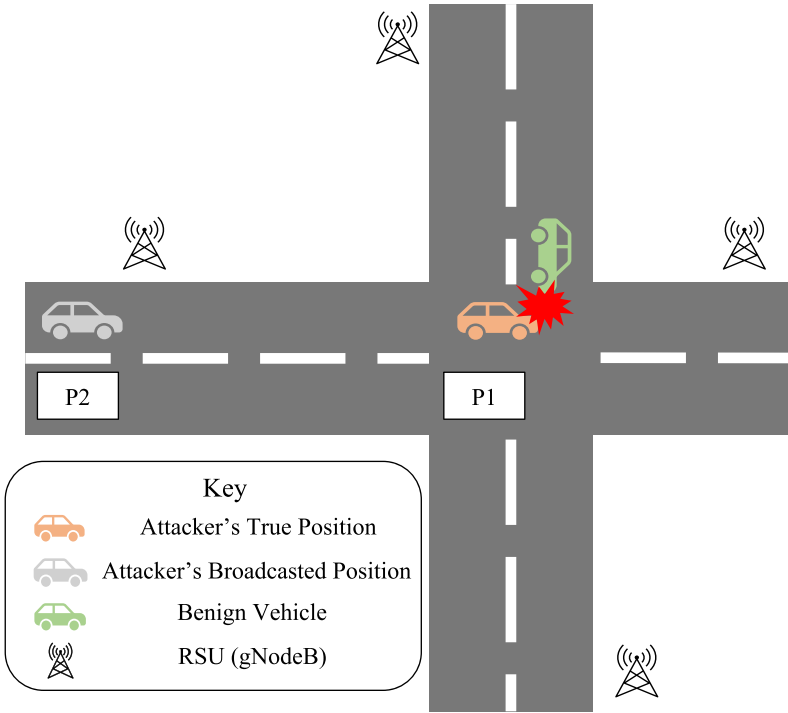


Fig. 2. The threat model under study.

Table 1
Description of attacks in VeReMi dataset.

ID	Attack	Description
1	Constant	$x = 5560, y = 5820$
2	Constant Offset	$\Delta x = 250, \Delta y = -150$
4	Random	Uniformly random in the playground
8	Random Offset	$\Delta x, \Delta y$ are uniformly random from $[-300, 300]$
16	Eventual Stop	Stop probability increases by 0.025 after each position update

3. Dataset description and pre-processing

The publicly available VeReMi Dataset (van der Heijden et al., 2018) is used to analyze the performance of NPFADS. The dataset contains samples of five position falsification attacks — constant, constant offset, random, random offset, and eventual stop. The description of the attacks is given in Table 1.

The attacker broadcasts a constant position of (5560,5820) in Type 1 attacks. In Type 2 attacks, the attacker adds a constant value of 250 on the x -axis and -150 on the y -axis. The attacker broadcasts random positions in Type 4 attacks, and in Type 8 attacks, the attacker still broadcasts random positions but constraints the range within $[-300, 300]$. In Type 16 attacks, the attacker suddenly starts to broadcast a fixed position with a probability of 0.025. During simulation, each vehicle generates a reception log containing periodic position updates from GPS and all received messages (BSMs from other vehicles) during simulation. Each log entry contains a reception time stamp, the claimed transmission time, the claimed sender, a simulation-wide unique message ID, a position vector, a speed vector, the RSSI, a position noise vector, and a speed noise vector. In addition, a ground truth file is updated whenever a message is sent by any vehicle, which contains the transmission time, sender, attacker type, message ID, and actual position/speed vectors.

For each simulation run, the entries in the reception logs are split into GPS and BSM messages. The GPS entries are discarded and are not used for further processing. The BSM messages from a single simulation run are aggregated, and duplicate records of a BSM message are removed. The position and speed vectors are split into individual fields (E.g., The Pos field is split into XPos, YPos, and ZPos). The fields 'ZPos', 'ZSpeed', 'pos_noise', and 'spd_noise' are removed as they are zero variance features. The 'RSSI' field is removed from the dataset to prevent the machine learning model from being dependent on the RSSI distribution. The fields 'type', 'rcvTime', and 'messageID' are also removed as they do not contribute to detecting position falsification attacks. Two additional features are calculated for each BSM, 'XAcc' and 'YAcc', the acceleration in the X and Y direction, respectively. After pre-processing, each BSM contains the fields 'SendTime', 'XPos', 'YPos', 'XSpd', 'YSpd', 'XAcc', and 'YAcc'. For each unique sender in the dataset, a mobility matrix, M_i given by Eq. (1) (Gu et al., 2016) is constructed.

$$M_i = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} & m_{15} & m_{16} & m_{17} \\ m_{21} & m_{22} & m_{23} & m_{24} & m_{25} & m_{26} & m_{27} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ m_{n1} & m_{n2} & m_{n3} & m_{n4} & m_{n5} & m_{n6} & m_{n7} \end{bmatrix} \quad (1)$$

where m_{n1} is the time, $m_{n2}, m_{n3}, m_{n4}, m_{n5}, m_{n6}$ and m_{n7} corresponds to XPos, YPos, XSpd, YSpd, XAcc, and YAcc at the time 'n'. The mobility matrix, M_i is then centralized, i.e., the mean of the column is subtracted from each entry in the column to ensure that the matrix's eigenvalues are not complex (Quevedo et al., 2020). For each mobility matrix M_i , the transpose, M_i^T , is calculated. M_i is multiplied with the M_i^T to generate a square matrix, M_i^S as shown in Eq. (2).

$$M_i^S = M_i \times M_i^T \quad (2)$$

The eigenvalues of the square matrix M_i^S are computed. The above computations represent each vehicle's physical movement in the simulation as a set of seven features. The attack label is appended towards the end of the seven eigenvalues to create a labeled dataset.

4. Novel Position Falsification Attack Detection System for the Internet of Vehicles (NPFADS for the IoV)

Readers may refer to Fig. 1 for a clear understanding of the placement of various components of NPFADS that are explained in this section. The proposed NPFADS is shown in Fig. 3. The NPFADS consists of two modules — the Misbehavior Detection System (MDS) and the Novel Attack Detection Module (NADM). The MDS protects the vehicles in the network from position falsification attacks. The NADM identifies signatures of any new position falsification attacks emerging in the network and extracts samples of the new attack to retrain the MDS. The MDS is installed in each OBU and fog node. The MDS at the OBU protects individual vehicles, whereas the MDS at the fog node protects all the vehicles in the network from position falsification attacks. The NADM is installed only in the fog nodes. The fog nodes are connected to the cloud server, which holds a copy of both the MDS and the NADM.

The OBU in each vehicle maintains a local database of BSMs, the OBU-BSM Database (OBU-BSMD), that stores the BSMs received from other vehicles within the DSRC communication range. The fog nodes also hold a database of BSMs received from each vehicle within its scope, the Fog Node-BSM Database (FN-BSMD). If the vehicle or the fog node does not receive any BSM from a particular vehicle for 100 s, the records of that vehicle are deleted from the respective databases assuming that the sender is beyond the communication range. The fog nodes also hold a shared database, Misbehaving Vehicles Database (MVD), that contains the list of misbehaving vehicles in the network. Each vehicle also stores a local copy of MVD in its OBU.

The OBU, on receiving a BSM, first performs misbehavior detection using an entity-centric approach by checking the MVD to see if the sender is blacklisted. If yes, the BSM is immediately discarded. If not, the BSM is added to OBU-BSMD. The MDS then performs misbehavior detection using a data-centric approach. The decision made using the data-centric approach always overrules the entity-centric approach to ensure that any benign vehicle that has recently been compromised does not affect other vehicles in the network. By deciding on the received BSMs locally, the vehicle need not rely on infrastructure support for decision-making. The above-described mechanism also ensures that the vehicles remain protected in the event of loss of connection with the infrastructure or if the vehicle operates in environments that lack infrastructure.

The RSUs, upon receiving a BSM, immediately share the BSM with the fog nodes, which adds the BSM to the FN-BSMD. The misbehavior detection on the FN-BSMD using the MDS at the fog node is performed every 100 s. On detecting an attack, the fog node adds the sender's vehicle ID to the MVD. In the event of any update to the MVD, all fog nodes immediately download a copy of the MVD and distribute it to the vehicles connected to them through the RSU. The above-described mechanism ensures that a vehicle that is not in the vicinity of the attacker is also aware of the attack happening elsewhere in the system. Also, by performing the novel attack detection at the fog nodes, the amount of data traveling up the network is reduced. The vehicle flagged as misbehaving has to re-register itself with CA to further participate in the network. On re-assessing the flagged vehicle, the CA removes it from the MVD.

To check if any new position falsification attack has emerged in the network, the FN-BSMD is processed using NADM at the fog node. It is assumed that only one class of novel attacks emerge at a given time instant. If any NADM detects a novel class of attack, they extract the

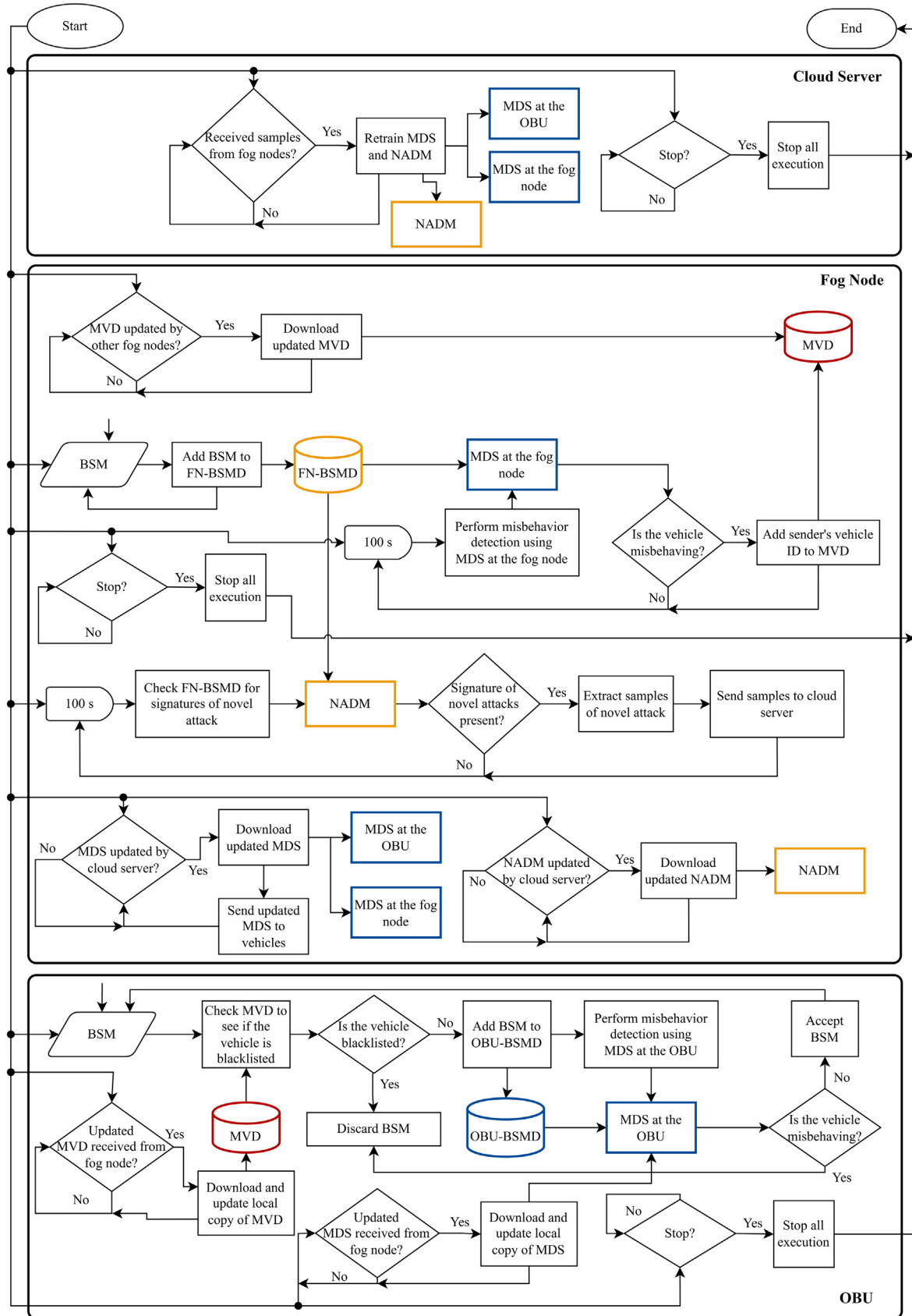


Fig. 3. The NPFADS.

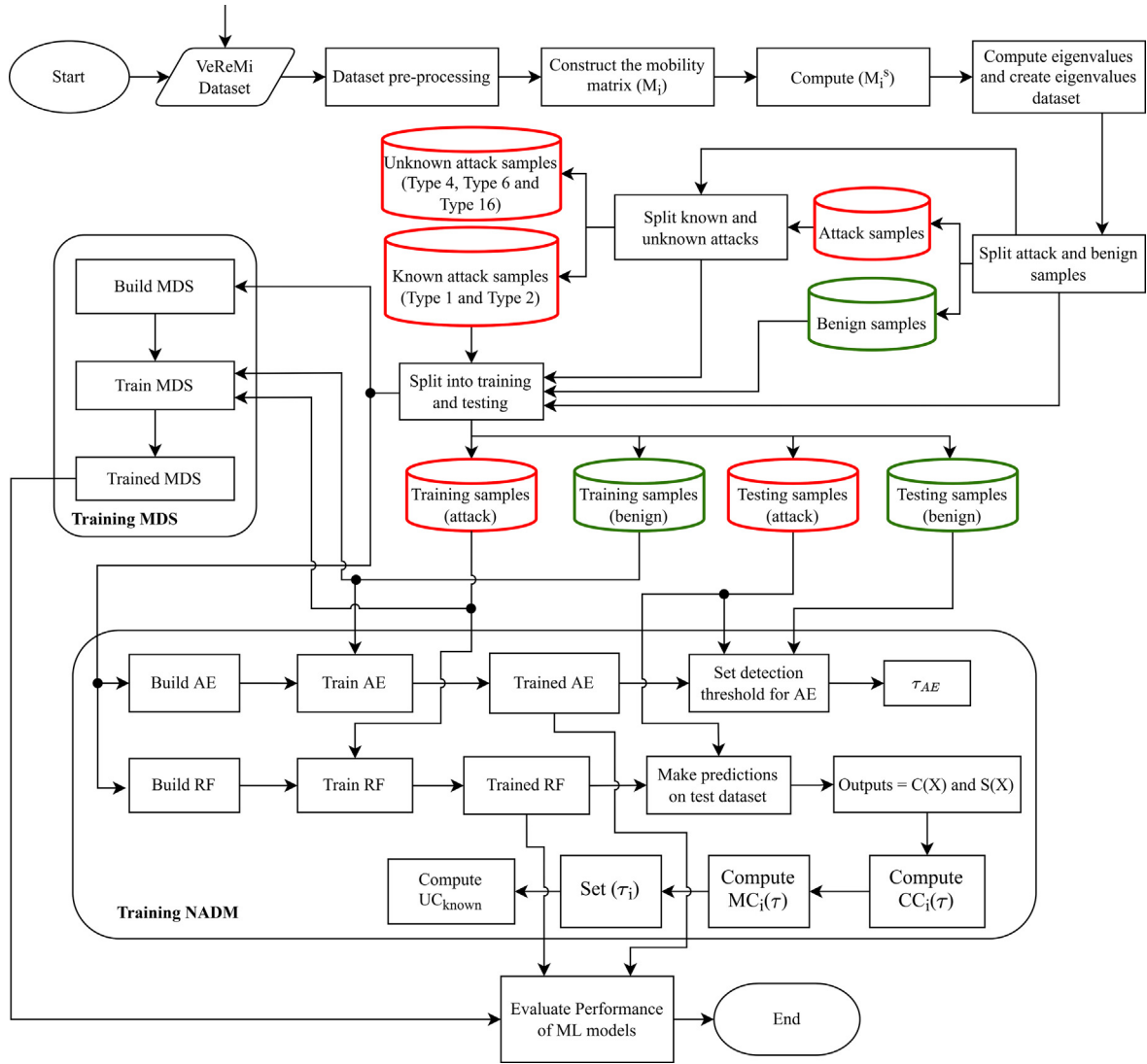


Fig. 4. NPFADS Initialization and Training Framework.

attack samples from FN-BSMD and share them with the cloud server. The cloud server retrain the MDS and NADM using the extracted samples and stores an updated copy. The fog nodes are programmed to periodically check the cloud server for updates and download the updated MDS and NADM. The updated MDS is sent to the OBUs through the RSUs. After the vehicles receive this updated information, they update their onboard MDS. The above-described mechanism ensures that the vehicles in the network are quickly immunized to the new position falsification attack in the network. The following subsections discuss the initialization and training framework of MDS and NADM in detail.

4.1. NPFADS initialization and training framework

Fig. 4 shows the NPFADS initialization and training framework. The working of NPFADS is described in time instances to keep track of the attack detection process. At time instant 1, the VeReMi dataset is pre-processed, and the eigenvalues dataset is divided into the attack and benign subsets. The attack dataset is split into known and unknown attack datasets. At this instant, it is assumed that only Type 1 and Type 2 attacks are known to the system and all other attacks (Type 4, Type 8 and Type 16) are considered novel. The datasets are then split into corresponding training and testing subsamples.

4.1.1. Initializing and training the MDS

A Random Forest (RF) model is used to build the MDS at the OBU and the fog nodes. The MDS is trained at the cloud server and is distributed to the vehicles and the fog nodes. A combination of the benign and known attack training samples is used to train the MDS. The hyperparameters were optimized using the Random Search with Cross-Validation library of scikit learn (Pedregosa et al., 2011). The performance of the RF models is then evaluated using the corresponding testing subsets, and the results are discussed in Section 5.

4.1.2. Initializing and training the NADM

The NADM consists of an AutoEncoder (AE) and a RF model. AE is a type of unsupervised machine learning algorithm used for anomaly detection. To detect signatures of a novel attack, the FN-BSMD is first passed through the AE to filter out the benign samples from the malicious samples. The malicious samples are then given as inputs to the RF model for further processing. The following subsections discuss the working of AE and RF in detail.

4.1.2.1. AE. The AE consists of an encoder and a decoder which are symmetrical about the bottleneck layer. The encoder maps an input 'x' onto a low-dimensional feature space, and the decoder aims to reconstruct the original inputs from the projected low-dimensional space. The aim is to minimize the reconstruction error between the

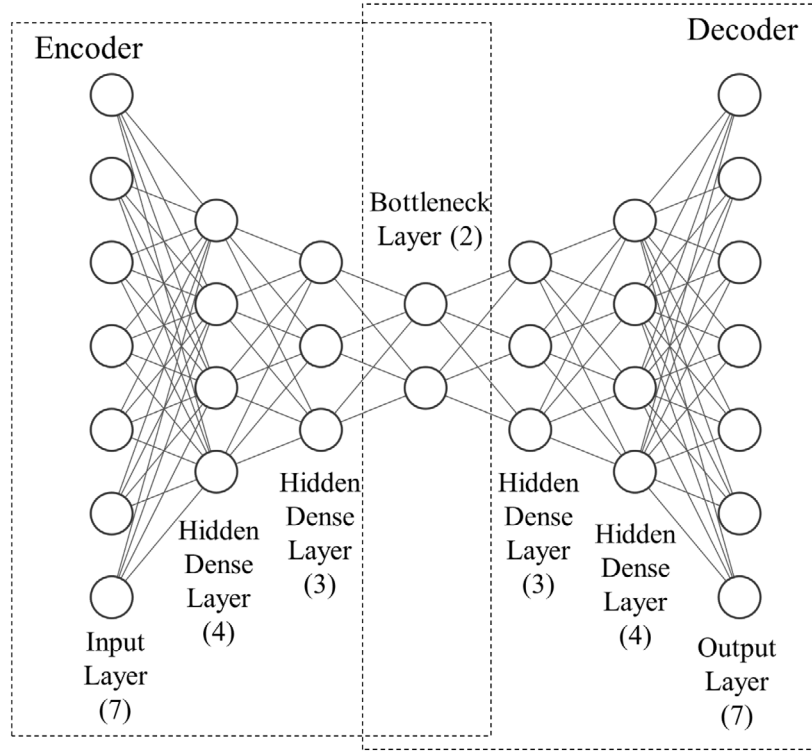


Fig. 5. AE model at NADM.

input x and the reconstruction $\hat{x}$. Fig. 5 shows the AE model used in the NADM. The input layer of the AE consists of 7 neurons, and the subsequent hidden layers contain 4 and 3 neurons, respectively. The bottleneck layer consists of two neurons, and the encoder's structure is mirrored for the decoder. In NADM, the AE is trained only on benign samples. Once trained, the AE will have minimal reconstruction error for benign samples and a higher reconstruction error for attack samples. This property of AE is exploited to filter the attack samples from benign samples. The Mean Squared Error (MSE) is used to measure the reconstruction error between x and $\hat{x}$. The threshold τ_{AE} is used to classify an input X with a reconstruction error MSE_X as benign when $MSE_X < \tau_{AE}$ or as malicious when $MSE_X \geq \tau_{AE}$.

The AE is trained using an Adam optimizer for 20 epochs with a batch size of 64. After each epoch, the benign testing samples is used to compute the validation loss. The training is stopped whenever the validation loss remains the same for three continuous epochs. To set τ_{AE} , the benign and the known attack testing samples are given as inputs to the AE, and the values of MSE are computed. The Threshold τ_{AE} is set to a MSE value that results in the best possible classification of benign and attack samples.

4.1.2.2. RF. The RF at NADM is trained in a supervised way using the set of known attacks. Once trained, for a given input X , the RF outputs a predicted class $C(X)$ with the highest prediction probability $S(X)$. When the input belongs to a known class of attack, the probability $S(X)$, that the prediction belongs to class $C(X)$, will typically be closer to 1. Whereas, when the given input belongs to a novel class, the RF will still try to predict a class within the set of known classes, but the probability $S(X)$ that the prediction belongs to class $C(X)$ will be lower. This property of supervised learning algorithms is exploited to detect novel position falsification attacks. For each unique class i in the set of known attacks, the Correct Classification Rate (CCR) and Misclassification Rate (MCR) defined by Eq. (3), and Eq. (4) (Stefano et al., 2000; Villon et al., 2020) are computed.

$$CCC_i(\tau) = \frac{\#((C(X) = i \text{ AND } S(X) > \tau) \text{ AND } Y = i)}{\#Y = i} \quad (3)$$

$$MC_i(\tau) = \frac{\#((C(X) \neq i \text{ AND } S(X) > \tau) \text{ AND } Y = i)}{\#Y = i} \quad (4)$$

where CCR_i is the correct classification rate for class i , MCR_i is the misclassification rate for class i , τ is the threshold, and Y is the true class. The values of $CCR_i(\tau)$ and $MCR_i(\tau)$ are computed iteratively by varying the value of τ from 0 to 1 in increments of 0.001. Using the values $CCR_i(\tau)$ and $MCR_i(\tau)$, a threshold τ_i is set for each class of known attacks. A sample is decided to belong to a known class if $S(X) \geq \tau_i$ and belongs to a novel class if $S(X) < \tau_i$. However, there may be samples of known attacks for which $S(X) < \tau_i$. To quantify the above phenomenon, the Unsure Classification rate for known attacks, UC_{known} is computed using Eq. (5), where N is the size of the dataset. UC_{known} is the percentage of predictions that are lower than the threshold τ_i for each known class of attack.

$$UC = \frac{\#((C(X) = i \text{ OR } C(X) \neq i) \text{ AND } S(X) < \tau_i)}{N} \quad (5)$$

To detect the presence of novel attacks, the FN-BSMD is passed through the AE. The marked malicious samples are used to compute the unsure classification rate of FN-BSMD, $UC_{FN-BSMD}$, using Eq. (5) and is compared with UC_{known} . If $UC_{FN-BSMD} > UC_{known}$, samples of a novel class of attacks exist in FN-BSMD. On detecting the presence of novel classes of attacks, the system extracts samples from the dataset where $S(X) < \tau_i$, i.e., samples for which the prediction probability is less than the detection threshold for the predicted class, $C(X)$. The system assigns a new class to the extracted samples, and the samples are added to the known attacks dataset. The RF at NADM and the MDS is retrained at the cloud server using these new samples.

First, to set the threshold τ_i for each class of known attacks using $CCC_i(\tau)$ and $MC_i(\tau)$, three hypotheses are proposed.

Hypothesis 1 (H1) - For each class, extract the thresholds for which the $CCR_i(\tau)$ is maximum. For these extracted thresholds, compute $MCR_i(\tau)$ and set τ_i to the value that results in the minimum misclassification rate. In the presence of multiple thresholds for which $MCR_i(\tau)$ is minimum, τ_i is set to the maximum value of τ for which $MCR_i(\tau)$ is minimum. H1 is mathematically represented in Eq. (6) and Eq. (7).

$$S_\tau = \arg_{\tau} \max (CCR_i(\tau)) \quad (6)$$

Table 2

Performance comparison of the hypotheses taken for the study.

Hypothesis	τ_1	τ_2	UC_{known}	$UC_{FN-BSMD}$	Detected the Presence of Novel Attacks?	NASEA
H1	0.539	0.589	0.00093	0.00219	Yes	0.00225
H2	0.049	0.469	0	0	No	–
H3	0.989	0.999	0.09617	0.99944	Yes	0.99944

$$\tau_i = \arg_{\tau' \in S_\tau} \min(MCR_i(\tau')) \quad (7)$$

Hypothesis 2 (H2) - For each class, extract the thresholds for which the $MCR_i(\tau)$ is minimum. For these extracted thresholds, compute the $CCR_i(\tau)$ and set τ_i to the value that results in the maximum value of correct classification rate. In the presence of multiple thresholds for which $CCR_i(\tau)$ is maximum, τ_i is set to the maximum value of τ for which $CCR_i(\tau)$ is maximum. H2 is mathematically represented in Eq. (8) and Eq. (9).

$$S_\tau = \arg_{\tau} \min(MCR_i(\tau)) \quad (8)$$

$$\tau_i = \arg_{\tau' \in S_\tau} \max(CCR_i(\tau')) \quad (9)$$

Hypothesis 3 (H3) - For each class, extract thresholds for which $CCR_i > 0.9$. For this subset of thresholds, the value of $MCR_i(\tau)$ is computed. The threshold for which the $MCR_i(\tau)$ is the minimum is set as the value of τ_i . H3 is mathematically represented in Eq. (10) and Eq. (11).

$$S_\tau = \arg_{\tau} CCR_i(\tau) > 0.90 \quad (10)$$

$$\tau_i = \arg_{\tau' \in S_\tau} \min(MCR_i(\tau')) \quad (11)$$

To quantitatively analyze the effectiveness of these hypotheses in setting τ_i , the thresholds τ_1 and τ_2 for Type 1 and Type 2 attacks are set using each of the above-described hypotheses. The corresponding value of UC_{known} is also computed. Since the system was not trained on Type 4 attacks, it is considered as a novel attack at this instant. The FN-BSMD dataset containing samples of Type 4 attacks and benign samples are given as inputs to the AE. The samples that are marked as malicious by the AE are given as inputs to the RF. The outputs $C(X)$ and $S(X)$ are used to compute the value of $UC_{FN-BSMD}$. Next, the attack detection ability of UC_{known} derived using each of the above-described hypotheses are assessed. If $UC_{FN-BSMD} > UC_{known}$, then the hypothesis detects the presence of a novel class. In NPFADS, on detecting the presence of a novel class, the samples for which $S(X) < \tau_i$ needs to be extracted. So, to quantify the number of novel attack samples that a hypothesis can extract, we define the metric, Novel Attack Sample Extraction Ability (NASEA), given in Eq. (12). NASEA is the percentage of novel attack samples in the dataset that can be extracted for further processing. The higher the extraction ability, the higher the number of novel attack samples extracted. This leads to better retraining of the MDS, thereby improving the attack detection performance. Table 2 shows the values of the thresholds (τ_1 and τ_2), the unsure classification rates and NASEA were derived using each of the above-described hypotheses.

Novel Attack Samples Extraction Ability (NASEA)

$$= \frac{\#S(X) < \tau_i}{\#Novel \text{ Attack Samples}} \quad (12)$$

Table 2 shows that the thresholds set using H1 and H2 are lower than those set using H3 because H1 and H2 prioritized achieving the best possible classification accuracy for known attack samples. This also resulted in H1 and H2 achieving a UC_{known} value of less than 1 percent, implying that there are very few samples for which $S(X) < \tau_i$. In contrast, H3 achieved an unsure classification rate of almost 10 percent by setting a relatively high threshold, thereby compromising the ability to classify known attacks perfectly. However, from the metric NASEA, it is evident that H3 can extract a higher number of samples than H1. H2 is unable to detect the presence of novel attacks and cannot be used to set detection thresholds. From the above analysis, it is concluded that H3 will be used to set the value of τ_i . Using H3, the value of τ_1 and τ_2 are set to 0.989 and 0.999, respectively.

Table 3

Demonstration of novel position falsification attack detection — RF at the NADM detection thresholds.

Time Instant	Class	Threshold (τ_i)
1	Type 1	0.989
	Type 2	0.985
3	Type 1	0.989
	Type 2	0.999
	Type 4	0.950

Table 4

Demonstration of novel position falsification attack detection — RF at the NADM outputs and decisions.

Time Instant	Sample	$S(X)$	$C(X)$	Decision
2	A	0.99	Type 1	Type 1 attack
	B	0.99	Type 2	Type 2 attack
	C	0.60	Type 1	Novel class of attack (Extract sample)
	D	0.80	Type 2	Novel Class of attack (Extract sample)
4	A	0.99	Type 1	Type 1 attack
	B	0.99	Type 2	Type 2 attack
	C	0.97	Type 4	Type 4 attack
	D	0.88	Type 4	Novel Class of attack (Extract sample)
	E	0.65	Type 1	Novel class of attack (Extract sample)

4.2. Case study

This subsection presents a case study to better help readers understand the novel attack detection process. At time instant 1, the system knows two position falsification attacks, Type 1 and Type 2. Let the thresholds τ_1 and τ_2 for the two known classes of attacks be 0.989 and 0.985 (shown in Table 3), and let UC_{known} be 0.09617. Once the system is initialized, the samples of Type 4 attacks (a novel attack at this instant) are introduced into the system. At time instant 2, the FN-BSMD containing samples of Type 1, Type 2, and Type 4 attacks are considered for the novel position falsification attack detection process. The FN-BSMD is passed through the AE, and the malicious samples are given as inputs to the RF. Let us assume that the AE marked four samples - A, B, C, and D as malicious. For each sample, the RF outputs $C(X)$ and $S(X)$ (shown in Table 4) and let the value of $UC_{FN-BSMD}$ be 0.10501. Since $UC_{FN-BSMD} > UC_{known}$, the presence of a novel class of attack is detected. The RF model predicted that sample A belonged to the Type 1 attack, sample B belonged to the Type 2 attack, sample C belonged to the Type 1 attack, and sample D belonged to the Type 2 attack with probabilities 0.99, 0.99, 0.6, and 0.8 respectively. However, by comparing $S(X)$ with the corresponding thresholds τ_1 and τ_2 , we observe the following: (i) It is evident that $S(A) > \tau_1$ and $S(B) > \tau_2$. Hence, we can conclude that the samples A & B are samples of a known attack. (ii) For samples C & D, $S(C) < \tau_1$ and $S(D) < \tau_2$. Therefore, the samples C & D belong to a novel attack.

At time instant 3, attack samples for which $S(X) < \tau_i$ are extracted and are used to retrain the MDS and the RF at the NADM. The thresholds, τ_1 , τ_2 and τ_3 , and the value of UC_{known} are computed. At this instant Type 4 attack is considered a known attack since NPFADS has learned to detect Type 4 attacks. The value of thresholds τ_1 , τ_2 and τ_3 are shown in Table 3. At time instant 4, a new FN-BSMD containing Type 1, Type 2, Type 4, and Type 8 attacks are considered for the novel position falsification attack detection process. At this instant, Type 8 attacks are considered to be novel attacks. The FN-BSMD is passed through the AE, and the five malicious samples - A, B, C, D, and E

Table 5

Performance metrics of NPFADS to known attacks.

Time Instant	Machine Learning Model	Class	Accuracy	Precision	Recall	F1 Score	Detection Time (μ s)
1	MDS at the OBU	Benign	–	0.97	1.00	0.98	20.31
		Type 1		0.99	0.98	0.98	
		Type 2		0.85	0.34	0.49	
		Attack		0.97	0.66	0.78	
	MDS at the fog node	Benign	–	0.97	0.99	0.98	21.09
		Type 1		0.99	1.00	1.00	
		Type 2		0.85	0.65	0.74	
		Attack		0.97	0.93	0.82	

Table 6

Performance metrics of NPFADS to novel attacks.

Time Instant	Machine Learning Model	Class	Accuracy	Precision	Recall	F1 Score	Detection Time (μ s)
2	MDS at the OBU	Benign	–	0.97	1.00	0.98	19.34
		Type 1		0.99	0.97	0.98	
		Type 2		0.84	0.34	0.48	
		Type 4		1.00	1.00	1.0	
		Attack		0.97	0.77	0.86	
	MDS at the fog node	Benign	–	0.97	0.99	0.98	22.13
		Type 1		0.98	1.00	0.99	
		Type 2		0.84	0.66	0.74	
		Type 4		1.00	1.00	1.00	
3	MDS at the OBU	Benign	–	0.97	1.00	0.98	20.00
		Type 1		0.99	0.97	0.98	
		Type 2		0.84	0.34	0.48	
		Type 4		1.00	1.00	1.00	
		Type 8		1.00	1.00	1.00	
		Attack		0.97	0.83	0.90	
	MDS at the fog node	Benign	–	0.97	0.99	0.98	21.80
		Type 1		0.99	1.00	0.99	
		Type 2		0.84	0.66	0.74	
		Type 4		0.99	1.00	1.00	
4	MDS at the OBU	Type 8	–	1.00	0.99	1.00	21.66
		Type 16		0.84	0.68	0.75	
		Attack		0.97	0.98	0.91	
		Attack		0.97	0.96	0.94	
	MDS at the fog node	Benign	–	0.97	0.98	0.98	23.49
		Type 1		0.99	1.00	1.00	
		Type 2		0.81	0.68	0.73	
		Type 4		0.99	1.00	1.00	
		Type 8		1.00	1.00	1.00	
		Type 16		0.98	0.97	0.98	
		Attack		0.97	0.96	0.92	

(as shown in Table 4) are given as inputs to the RF at the NADM, and the outputs are $C(X)$ and $S(X)$. The presence of a novel attack is detected and by comparing $UC_{FN-BSM D}$ with UC_{known} . By comparing $S(X)$ with the corresponding thresholds, it is concluded that A, B, and C are samples of a known attack, and C & D are samples of a novel attack. The above-described process is repeated to detect novel attacks emerging in the network.

5. Performance analysis

Given that all the modules of NPFADS have been trained and initialized, we now analyze the attack detection performance. The performance metrics used to analyze the machine learning models are defined in Eq. (13)–(16). Table 5 shows the performance of NPFADS at time instant 1. The primary objective of the MDS is to protect the vehicles from position falsification attacks irrespective of the type of

attacks. So, the performance metrics of binary classification have been evaluated by setting $C(X) = 1$ where $C(X) \neq 0$ and are presented in Table 5 under the class ‘Attack’. However, the individual classification performance has also been discussed for a comprehensive analysis.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (13)$$

$$Precision = \frac{TP}{TP + FP} \quad (14)$$

$$Recall = \frac{TP}{TP + FN} \quad (15)$$

$$F1 \text{ score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (16)$$

It is evident that the MDS at the OBU can detect Type 1 attacks with excellent accuracy indicated by the F1 scores in Table 5. But the MDS performs poorly in detecting Type 2 attacks. Type 2 attacks are constant offset attacks, by which an attacker adds a constant value to

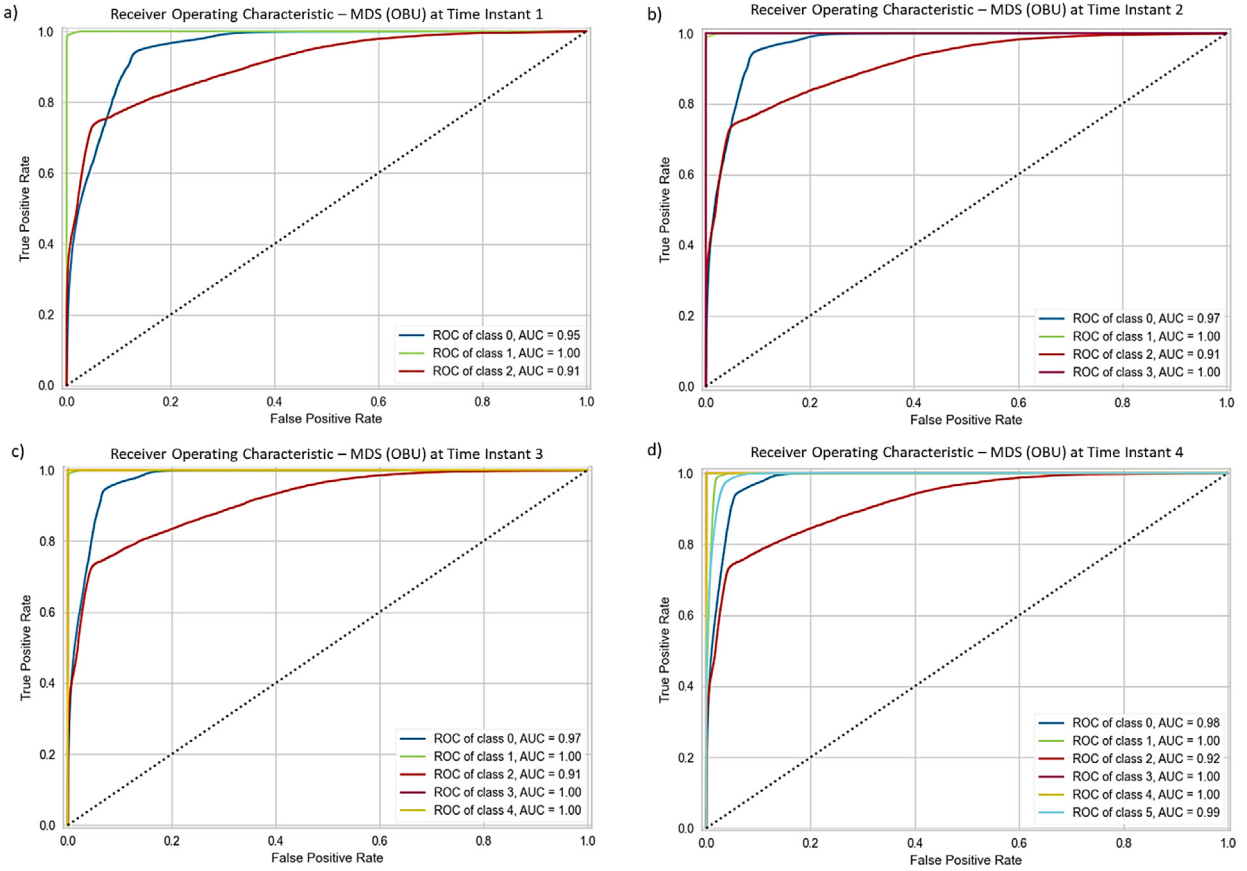


Fig. 6. ROC Curves of MDS at the OBU at (a) Time instant 1; (b) Time instant 2; (c) Time instant 3; (d) Time instant 4.

the actual position. So, the eigenvalues of Type 2 attacks are similar to the eigenvalues of benign samples. Hence, there is a performance drop when the model is trained and tested with a dataset containing both benign and Type 2 attack samples. The MDS at the fog nodes detects position falsification attacks with better accuracy than the MDS at the OBUs for all vehicle densities. The reason is that the fog nodes perform misbehavior detection over the aggregated messages from the RSUs every 100 s. Therefore, the MDS at the fog nodes hold more BSM samples to understand the vehicles' behavior before arriving at a decision. In contrast, the MDS at the OBUs performs misbehavior detection each time a BSM is received.

At time instant 2, samples of Type 4 attacks are introduced into the system. The RSUs share the BSM samples with the fog nodes, which create the FN-BSMD. The FN-BSMD is passed to the AE for anomaly detection. Samples of those with the MSE greater τ_{AE} will be marked as an anomaly and will be passed to the RF in the NADM. The RF in the NADM makes predictions on the samples and outputs $S(X)$. The unsure classification rate of the FN-BSMD is computed, and the presence of a novel class of attacks is detected. The NADM compares the prediction probabilities $S(X)$ with τ_i and extracts samples with $S(X) < \tau_i$. The extracted samples will be assigned a new class label and are used to retrain the NADM and the MDS at the cloud server. The performance metrics of the RF models for the new class of the attack are shown in Table 6. In subsequent time instances 3 and 4, Type 8 and Type 16 attacks are introduced to the system. A procedure similar to detecting Type 4 attacks will start. The performance metrics of the model in detecting Type 8 and Type 16 attacks are also shown in Table 6.

From Table 6, it is evident that the detection performance of the MDS at the OBUs on known attacks has not degraded when novel attack detection capabilities have been added. The empirical results show that at instant 2, the detection performance of Type 1 and Type 2 attacks has not degraded when Type 4 attack detection capability has been

added. A similar trend can be observed at time instant 3. However, at time instant 4, the performance of Type 1 attacks slightly degrades when Type 16 attack detection capability is added. This is because the behavior of Type 16 attacks is similar to Type 1 attacks. The attacker initially broadcasts the correct position and then starts to broadcast the same position after a specific period. From Tables 5 and 6, it is evident that the detection times of the ML models largely remain the same across all time instances for both the MDS at the OBU and the MDS at the fog node. The minor fluctuations in the detection times can be attributed to the environmental variations in the system that is used for testing.

The Receiver Operating Characteristic (ROC) Curves and the Precision–Recall (PR) Curves has been used to analyze the performance of the machine learning models. The ROC curves of MDS at the OBU and MDS at the fog node at time instances 1, 2, 3, and 4 are shown in Figs. 6 and 7, respectively. The PR Curves for the MDS at the OBU and the MDS at the fog node are shown in Fig. 8 and Fig. 9, respectively. The labels 'class 0', 'class 1', 'class 2', 'class 3', 'class 4' and 'class 5' in Fig. 6, Fig. 7, Figs. 8 and 9 correspond to the classes 'Benign', 'Type 1', 'Type 2', 'Type 4', 'Type 8', and 'Type 16' respectively.

The ROC Area Under the Curve (AUC) represents the degree of separability — the ability of a model to differentiate between classes. From Fig. 6, the ROC AUC of MDS at the OBU, for all position falsification attacks, is greater than 0.9, indicating that the model can differentiate between classes of position falsification attacks with high probability. The ROC AUC for Type 1, Type 4, and Type 8 attacks is 1, indicating that the MDS at the OBU can differentiate between these classes perfectly. Fig. 7 shows that the MDS at the fog node can perfectly distinguish between Type 1, Type 4, Type 8, and Type 16 attacks. The model's performance towards Type 2 attacks is equally good, which is evident from a ROC AUC value of 0.9. The PR curves quantify the tradeoff between precision and recall. The PR AUC summarizes this

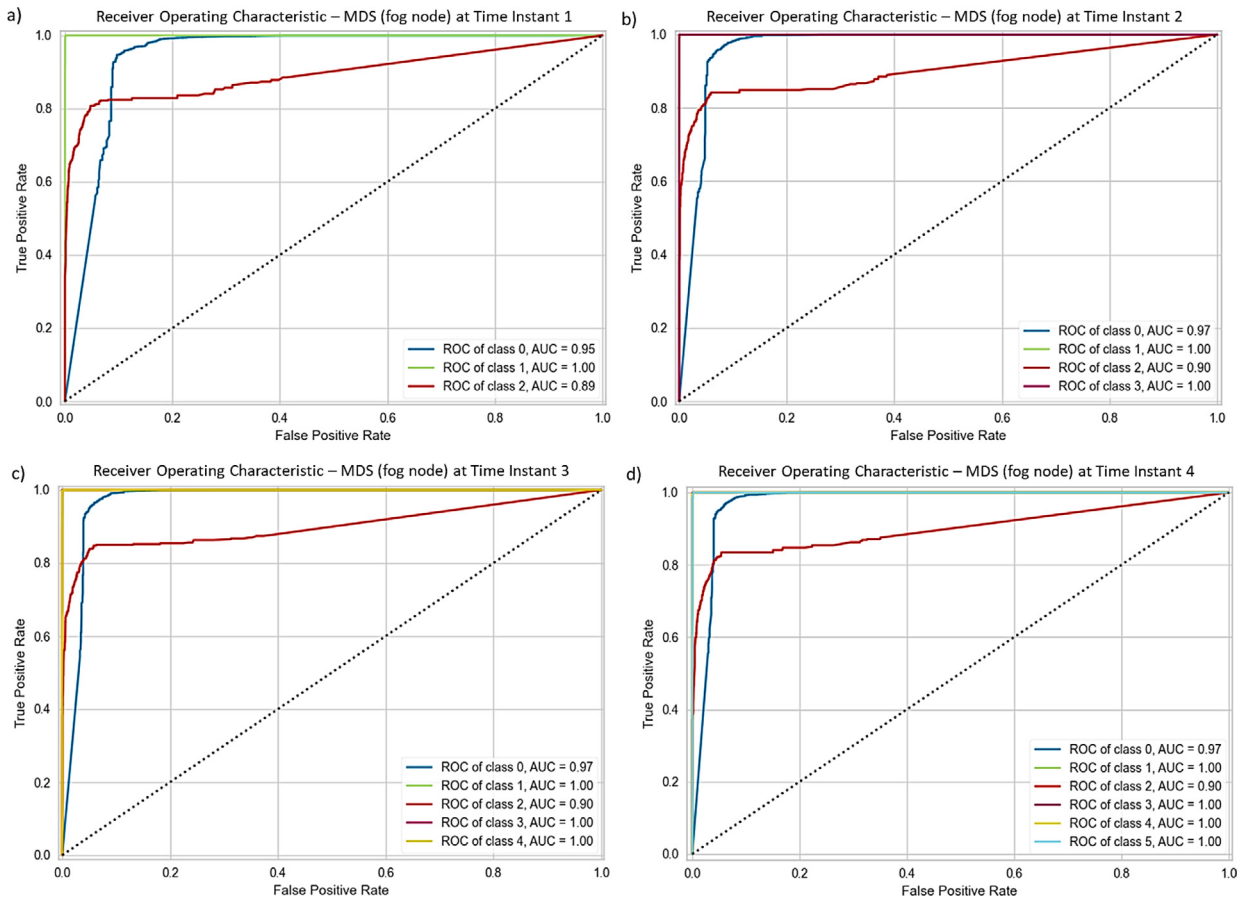


Fig. 7. ROC Curves of MDS at the fog node at (a) Time instant 1; (b) Time instant 2; (c) Time instant 3; (d) Time instant 4.

relationship. From Fig. 8, the PR AUC for MDS at the OBU shows good performance for Type 1, Type 4, and Type 8 attacks across time instances 1, 2, and 3. However, at time instant 4, performance towards Type 1 attacks degrades when Type 16 attack detection capability is added. In contrast, the performance of MDS at the fog node does not degrade across all time instances which is evident from Fig. 9. Overall, from the performance analysis, it is evident that the performance of the system in detecting position falsification attacks does not degrade when new attack detection capabilities are added. The analysis also shows that NPFADS can learn and detect novel position falsification attacks emerging in the network.

Table 7 compares the performance of NPFADS to existing MDSs in the literature. The model's performance towards individual classes is averaged to arrive at the overall performance metric for the MDS in Sharma and Jaekel (2022). It is evident that NPFADS outperforms the MDSs in Sharma and Liu (2021), So et al. (2019). The MDS in Sharma and Jaekel (2022) has better overall attack detection performance than our proposed model because the authors removed the benign samples in the dataset and used only the attack samples for performance evaluation. The comparative analysis shows that NPFADS learns and detects novel position falsification attacks and achieves attack detection performance that is on par with the existing supervised learning models in literature. Our proposed model achieves this good performance even when initialized with zero knowledge about the novel attacks (Type 4, Type 8 and Type 16).

6. Conclusion

In this work, we proposed a NPFADS for the IoV to detect novel position falsification attacks emerging in IoV networks. The working methodology of the system was described in detail, along with the novel

position falsification attack detection strategy. The performance of the system was analyzed using the metrics accuracy, precision, recall, F1 score, ROC and PR curves. The analysis showed that NPFADS could learn and detect emerging attacks in the network. The performance of the proposed systems was also compared to existing MDSs in the literature. The comparison has shown that the NPFADS outperforms the existing MDSs in the literature even when initialized with zero knowledge about the novel position falsification attacks.

CRediT authorship contribution statement

Harun Surej Ilango: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Writing – original draft, Visualization. **Maode Ma:** Validation, Writing – review and editing, Revision, Supervision. **Rong Su:** Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used are all shared in the article.

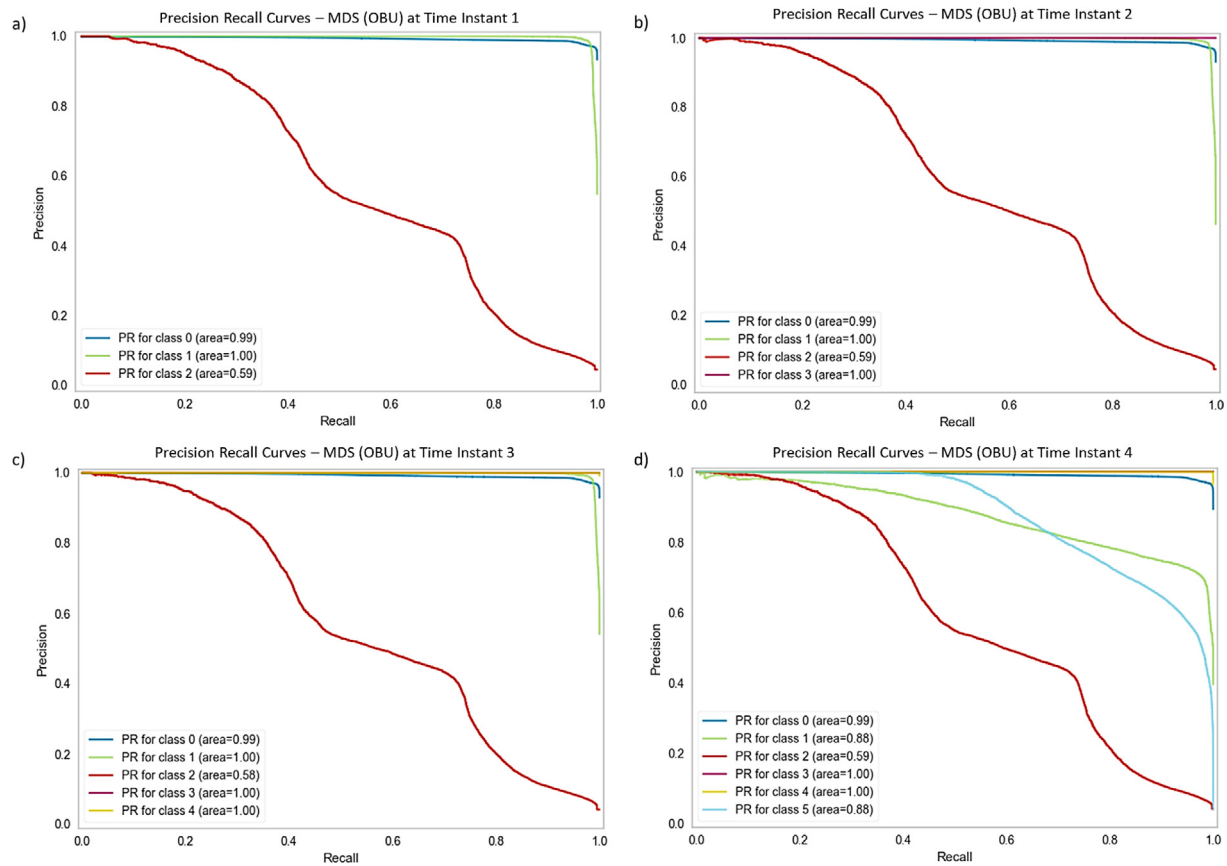


Fig. 8. PR Curves of MDS at the OBU at (a) Time instant 1; (b) Time instant 2; (c) Time instant 3; (d) Time instant 4.

Table 7

Performance comparison of NPFADS with existing works in literature.

Work	Model	Attack Type	Precision	Recall	F1 Score
Sharma and Liu (2021)	Ensemble	Type 1	1	0.8579	0.92
		Type 2	0.5455	0.0275	0.05
		Type 4	0.9997	0.8801	0.94
		Type 8	0.9218	0.7395	0.82
		Type 16	0.6388	0.4185	0.51
		Overall	0.887	0.5759	0.7
Sharma and Jaekel (2022)	KNN	Type 1	1.00	1.00	1.00
		Type 2	0.998	0.997	1.00
		Type 4	1.00	0.998	1.00
		Type 8	0.999	0.972	0.99
		Type 16	0.968	0.952	0.96
		Overall	0.981	0.985	0.98
So et al. (2019)	WBSM	Type 1	0.9545	0.9286	0.94
		Type 2	0.9424	0.7244	0.82
		Type 4	0.9681	1.00	0.98
		Type 8	0.9402	0.8049	0.87
		Type 16	0.9404	0.9404	0.94
		Overall	0.9591	0.8373	0.89
NPFADS	MDS at the OBU	Type 1	0.76	0.85	0.8
		Type 2	0.84	0.34	0.48
		Type 4	1.00	1.00	1.00
		Type 8	1.00	1.00	1.00
		Type 16	0.84	0.68	0.75
		Overall	0.98	0.85	0.91
	MDS at the fog node	Type 1	0.99	1.00	1.00
		Type 2	0.81	0.68	0.73
		Type 4	0.99	1.00	1.00
		Type 8	1.00	1.00	1.00
		Type 16	0.98	0.97	0.98
		Overall	0.96	0.92	0.94

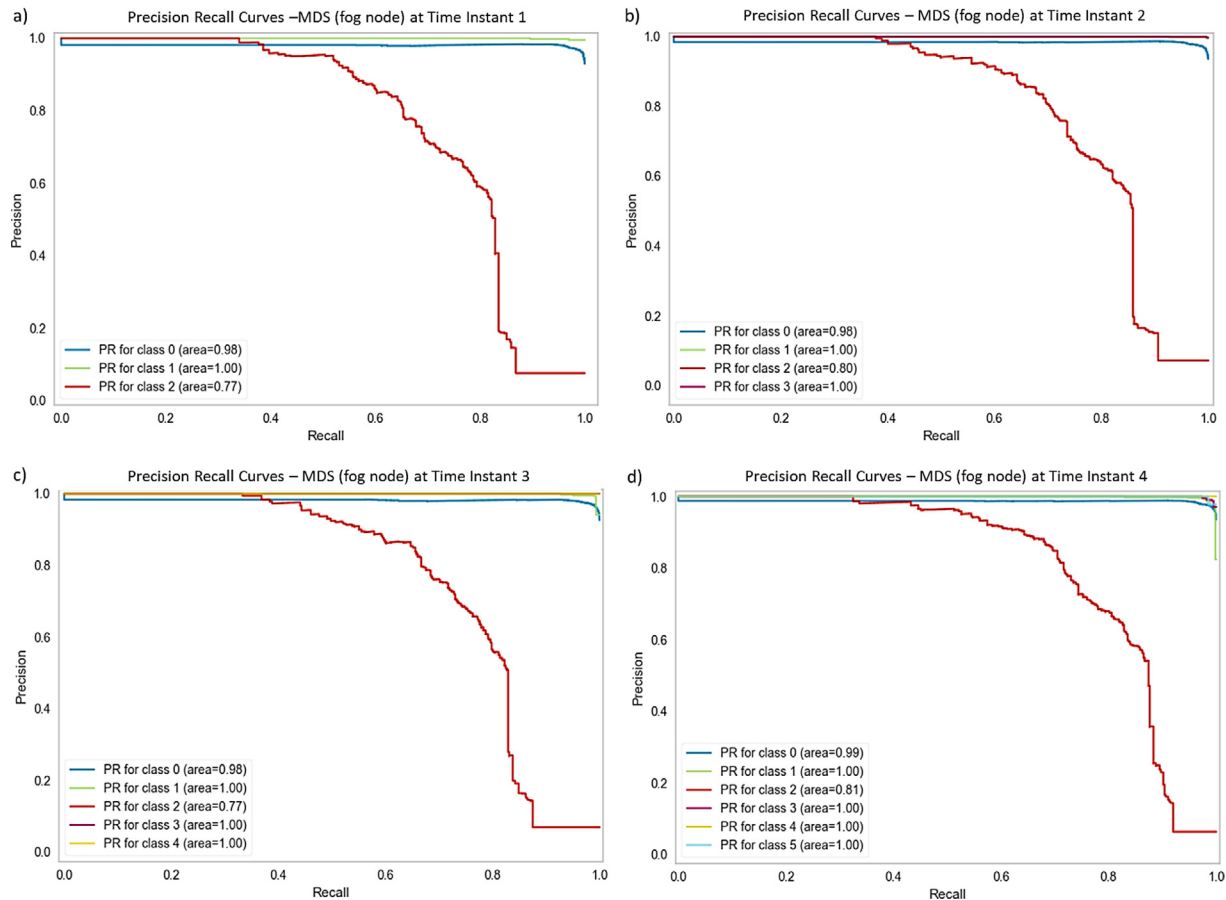


Fig. 9. PR Curves of MDS at the fog node at (a) Time instant 1; (b) Time instant 2; (c) Time instant 3; (d) Time instant 4.

Acknowledgment

This research is supported by A*STAR under its RIE2020 Advanced Manufacturing and Engineering (AME) Industry Alignment Fund – Pre Positioning (IAF-PP) (Award A19D6a0053). Any opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and do not reflect the views of A*STAR.

References

- Adat, V., Gupta, B.B., 2018. Security in internet of things: Issues, challenges, taxonomy, and architecture. *Telecommun. Syst.* 67 (3), 423–441. <http://dx.doi.org/10.1007/s11235-017-0345-9>, 2018.
- Ercan, S., Ayaida, M., Messai, N., 2022. Misbehavior detection for position falsification attacks in VANETs using machine learning. *IEEE Access* 10, 1893–1904. <http://dx.doi.org/10.1109/ACCESS.2021.3136706>.
- Gu, P., Khatoun, R., Begriche, Y., Serhrouchni, A., 2016. Vehicle driving pattern based sybil attack detection. In: 2016 IEEE 18th International Conference on High Performance Computing and Communications; IEEE 14th International Conference on Smart City; IEEE 2nd International Conference on Data Science and Systems. HPCC/SmartCity/DSS, 12–14 Dec. 2016, pp. 1282–1288. <http://dx.doi.org/10.1109/HPCC-SmartCity-DSS.2016.0182>.
- Kamal, M., Srivastava, G., Tariq, M., 2021. Blockchain-based lightweight and secured V2V communication in the internet of vehicles. *IEEE Trans. Intell. Transp. Syst.* 22 (7), 3997–4004. <http://dx.doi.org/10.1109/TITS.2020.3002462>.
- Katiyar, P., Kumarn, U., Balakrishnan, S., 2013. Detection and discrimination of DDoS attacks from flash crowd using entropy variations. *Int. J. Eng. Technol.* 5, 3514–3519, 08/01.
- Pedregosa, F., et al., 2011. Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Quevedo, C.H.O.O., Quevedo, A.M.B.C., Campos, G.A., Gomes, R.L., Celestino, J., Serhrouchni, A., 2020. An intelligent mechanism for sybil attacks detection in VANETs. In: ICC 2020-2020 IEEE International Conference on Communications. ICC, 7–11 2020, pp. 1–6. <http://dx.doi.org/10.1109/ICC40277.2020.9149371>.
- Sharma, A., Jaekel, A., 2022. Machine learning based misbehaviour detection in VANET using consecutive BSM approach. *IEEE Open J. Veh. Technol.* 3, 1–14. <http://dx.doi.org/10.1109/OJVT.2021.3138354>.
- Sharma, P., Liu, H., 2021. A machine-learning-based data-centric misbehavior detection model for internet of vehicles. *IEEE Internet Things J.* 8 (6), 4991–4999. <http://dx.doi.org/10.1109/JIOT.2020.3035035>.
- So, S., Petit, J., Starobinski, D., 2019. Physical layer plausibility checks for misbehavior detection in V2X networks. In: Presented at the Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks. Miami, Florida.
- Stefano, C.D., Sansone, C., Vento, M., 2000. To reject or not to reject: That is the question—an answer in case of neural classifiers. *IEEE Trans. Syst., Man, Cybern., Part C (Applications and Reviews)* 30 (1), 84–94. <http://dx.doi.org/10.1109/5326.827457>.
- van der Heijden, R.W., Lukaseder, T., Kargl, F., 2018. VeReMi: A Dataset for Comparable Evaluation of Misbehavior Detection in VANETs. Springer International Publishing, Cham, pp. 318–337. http://dx.doi.org/10.1007/978-3-030-01701-9_18, In Security and Privacy in Communication Networks.
- Villon, S., Mouillot, D., Chaumont, M., Subsol, G., Claverie, T., Villéger, S., 2020. A new method to control error rates in automated species identification with deep learning algorithms. *Sci. Rep.* 10 (1), 10972. <http://dx.doi.org/10.1038/s41598-020-67573-7>, 2020/07/03.
- Xu, X., et al., 2022. Service offloading with deep Q-network for digital twinning-empowered internet of vehicles in edge computing. *IEEE Trans. Ind. Inf.* 18 (2), 1414–1423. <http://dx.doi.org/10.1109/TII.2020.3040180>.